Algebra Exercises

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Winter Break, 2024/2025

CAS MA 741

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Chapter 8

Linear Algebra, Reprise

8.1 Functors

Exercise 8.1.1. Let $\mathscr{F} : \mathsf{C} \to \mathsf{D}$ be a covariant functor, and let C and D have products. Show there exists a unique morphism $\mathscr{F}(A \times B) \to \mathscr{F}(A) \times \mathscr{F}(B)$ making the relevant diagram commute.

If D has coproducts (denoted $A \amalg B$) and $\mathscr{G} : \mathsf{C} \to \mathsf{D}$ is contravariant, show there exists a unique morphism $\mathscr{G}(A) \amalg \mathscr{G}(B) \to \mathscr{G}(A \times B)$.

Solution. By definition of a functor, $\mathscr{F}(A \times B)$ is an object in D, and

$$\begin{tikzcd} & \mathscr{F}(A \times B) \arrow[dl, "\mathscr{F}(\pi_A)"'] \arrow[dr, "\mathscr{F}(\pi_B)"] & \\ \mathscr{F}(A) & & \mathscr{F}(B) \end{tikzcd}$$

is a diagram in D. Therefore, by universal property of products, the diagram

$$\begin{tikzcd} & & \mathscr{F}(A) \\ \mathscr{F}(A \times B) \arrow[r, dashed] \arrow[urr, bend left, "\mathscr{F}(\pi_A)"] \arrow[drr, bend right, "\mathscr{F}(\pi_B)"'] & \mathscr{F}(A) \times \mathscr{F}(B) \arrow[ur, "\pi_{\mathscr{F}(A)}"] \arrow[dr, "\pi_{\mathscr{F}(B)}"] & \\ & & \mathscr{F}(B) \end{tikzcd}$$

commutes.

Now suppose $\mathscr{G} : \mathsf{C} \to \mathsf{D}$ is contravariant and D has coproducts. Since there exist projections $\pi_A : A \times B \to A$, $\pi_B : A \times B \to B$ in C, and since $\mathscr{G}$ is contravariant, there exist morphisms

$$\mathscr{G}(\pi_A) : \mathscr{G}(A) \to \mathscr{G}(A \times B), \qquad \mathscr{G}(\pi_B) : \mathscr{G}(B) \to \mathscr{G}(A \times B)$$

in D. Since D has coproducts, by the universal property of coproducts,

$$\begin{tikzcd}
\mathscr{G}(A) \arrow[dr, "i_A"] \arrow[drr, bend left, "\mathscr{G}(\pi_A)"] & & \\
& \mathscr{G}(A) \amalg \mathscr{G}(B) \arrow[r, dashed] & \mathscr{G}(A \times B) \\
\mathscr{G}(B) \arrow[ur, "i_B"] \arrow[urr, bend right, "\mathscr{G}(\pi_B)"'] & &
\end{tikzcd}$$

exists in D and commutes. ■

Exercise 8.1.2. Let $\mathscr{F} : \mathsf{C} \to \mathsf{D}$ be a fully faithful functor. If A, B are in C, show that $A \cong B$ if and only if $\mathscr{F}(A) \cong \mathscr{F}(B)$.

Solution. trivial, do later ■

Exercise 8.1.3. Formalize the notion of a presheaf of abelian groups on a topological space T. If $\mathscr{F}$ is a presheaf on T, elements of $\mathscr{F}(U)$ are called *sections* of $\mathscr{F}$ on U. The homomorphism $\rho_{UV} : \mathscr{F}(U) \to \mathscr{F}(V)$ induced by an inclusion $V \subseteq U$ is called the *restriction map*.

Show that a presheaf is obtained by letting $\mathscr{C}(U)$ be the additive abelian group of complex valued functions on U, with restriction of sections defined by ordinary restriction of functions.

For this presheaf, prove that one can uniquely glue sections agreeing on overlapping sets. That is, if U, V are open sets and $s_U \in \mathscr{C}(U)$, $s_V \in \mathscr{C}(V)$ agree after restriction to $U \cap V$, prove that there exists a unique $s \in \mathscr{C}(U \cup V)$ such that s restricts to s_U on U and s_V on V.

Solution. Formalizing the notion is very simple. I will change notation slighty: let $(X, \mathcal{T})$ be a topological space, and define the category T such that $\mathrm{Obj}(\mathsf{T}) = \mathcal{T}$ and

$$\mathrm{Hom}_{\mathsf{T}}(U, V) = \begin{cases} \varnothing, & U \not\subseteq V \\ \{*\}, & U \subseteq V \end{cases}.$$

Then a presheaf $\mathscr{F}$ on X is a contravariant functor $\mathscr{F} : \mathsf{T} \to \mathsf{Ab}$.

Let $\mathscr{C} : \mathsf{T} \to \mathsf{Ab}$ be the contravariant assignment given by

$$\mathscr{C}(U) = (C(U, \mathbb{C}), +).$$

That is, U maps to the set of continuous functions $f : U \to \mathbb{C}$, viewed as a group under addition. Morphisms $i : U \hookrightarrow V$ are defined as

$$\mathscr{C}(i)(f) = \rho_{VU}(f) = f|_U.$$

That is, the restriction to U of a section f of $\mathscr{C}$ over V is the standard function restriction. Verification that the group given is abelian is immediate, so now we show that the restriction morphisms are group homomorphisms. Note that if $f, g \in \mathscr{C}(V)$ and $U \subseteq V$, then

$$\rho_{VU}(f+g)(x) = (f+g)|_U(x) \coloneqq f|_U(x) + g|_U(x) = \rho_{VU}(f)(x) + \rho_{VU}(g)(x).$$

Therefore, this is a group homomorphism by definition of the operation. Next, we show identity maps map to identity maps. Note that for any $f : U \to U$,

$$\rho_{UU}(f) = f_U = f.$$

Therefore, $\rho_{UU} = 1_{C(U)}$. Finally, we wish to show that if $U \subseteq V \subseteq W$, then $\rho_{WU} = \rho_{WV}\rho_{VU}$. Let $f : W \to \mathbb{C}$. We have

$$\rho_{WU}(f) = f_U, \qquad \rho_{WV}\rho_{VU}(f) = \rho_{VU} f_V = f_U.$$

Therefore, $\mathscr{C}$ is a contravariant functor, and thus a presheaf of abelian groups on X.

Finally, the gluing condition. Let $s_U \in \mathscr{C}(U)$ and $s_V \in \mathscr{C}(V)$ agree on $U \cap V$. We can define a function $s \in \mathscr{C}(U \cup V)$ as the map

$$s(x) = \begin{cases} s_U(x), & x \in U \\ s_V(x), & x \in V \end{cases}$$

.

Note that this is well-defined because if $x \in U$ and $x \in V$, then $s_U(x) = s_V(x)$. Moreover, it is manifestly unique since it is uniquely determined by s_U and s_V, and since they are continuous, s is clearly continuous as well due to the fact that s_U and s_V agree on $U \cap V$. Therefore, $\mathscr{C}$ is a sheaf of abelian groups on X. ■

Exercise 8.1.4. Verify that in R-Mod, the equalizer of $\phi : A \to B$ and $0 : A \to B$ is the kernel $\ker \phi$.

Solution. Verification is simple. Note that for all objects X, the diagram

$$\begin{tikzcd}
A \arrow[rr, bend left, "\phi"] \arrow[rr, bend right, "0"'] & & B \\
& \varprojlim \mathscr{F} \arrow[ul, "f_A"] \arrow[ur, "f_B"'] & \\
& X \arrow[u, dashed] \arrow[uul, bend left, "x_A"] \arrow[uur, bend right, "x_B"'] &
\end{tikzcd}$$

commutes. Moreover, $f_B = 0f_A = \phi f_A$. However, since $0f_A = 0$, we have $\phi f_A = 0$, and $f_B = 0$.

We also have $x_B = 0x_A = \phi x_A$, and thus $x_B = 0$. We can rewrite this diagram as

$$\begin{tikzcd}
X \arrow[r, "x_a"] \arrow[rr, bend left, "0"] \arrow[dr, dashed] & A \arrow[r, "\phi"] & B \\
& \varprojlim \mathscr{F} \arrow[u, "f_A"] \arrow[ur, "0"'] &
\end{tikzcd}$$

Note that this is precisely the universal property satisfied by the kernel, since by commutativity of the diagram, we have that $\phi f_A = 0$, and we can actually drop the 0 morphism $\varprojlim \mathscr{F} \to B$ while retaining precisely the same information. Therefore, the same proof given that kernels satisfy this universal property in R-Mod suffices here to show that $\varprojlim \mathscr{F}$ exists, and is the kernel. ■

Exercise 8.1.5. Let I be the category consisting of elements of $\mathbb{Z}^+$ as objects, and a unique morphism $i \to j$ if and only if $i \leq j$. Let $\mathscr{F} : \mathsf{I} \to R$-$\mathsf{Mod}$ be a functor, consisting of a choice of object A_i for each i, and a morphism $\phi_{ji} : A_i \to A_j$ for all i, j. Note that an element of

$$\prod_i A_i$$

is a sequence $(a_i)_{i>0}$. Call a sequence (a_i) coherent if $\phi_{i+1,i}(a_{i+1}) = a_i$ for all i. Let A be the submodule of $\prod_i A_i$ consisting of all coherent sequences. Let $\phi_i : A \to A_i$ be the restriction of the canonical projection $\pi_i : \prod_i A_i \to A_i$ to A. Show that A, together with the morphisms ϕ_i, satisfy

$$\varprojlim_i A_i = A.$$

Solution. First, I want to show A is truly an R-module. Let $(a_i), (b_i) \in A$. Then

$$(a_i) - (b_i) = (a_i - b_i).$$

Note that (with the mild abuse of notation), we have

$$\phi_{i+1,i}(a_{i+1} - b_{i+1}) = \phi_{i+1,i}(a_{i+1}) - \phi_{i+1,i}(b_{i+1}) = (a_i) - (b_i) = (a_i - b_i).$$

Additionally, for $r \in R$,

$$r(a_i) = (ra_i).$$

Thus,

$$\phi_{i+1,i}(ra_{i+1}) = r\phi_{i+1,i}(a_{i+1}) = r(a_i) = (ra_i).$$

Therefore, A is an R-module, as perscribed. Now we show the relevant diagram commutes. Let $n \in \mathbb{Z}^+$. We have $\phi_n(a_i) = a_n$. We also have

$$(\phi_{n+1,n}\phi_{n+1})(a_i) = \phi_{n+1,n}(a_{n+1}) = a_n.$$

Therefore, the diagram

$$\begin{tikzcd}
 & & A \arrow[dll, "\dots"'] \arrow[dl, "\phi_4"] \arrow[d, "\phi_3"] \arrow[dr, "\phi_2"] \arrow[drr, "\phi_1"] & & \\
\cdots \arrow[r, "\phi_{45}"] & A_4 \arrow[r, "\phi_{34}"] & A_3 \arrow[r, "\phi_{23}"] & A_2 \arrow[r, "\phi_{12}"] & A_1
\end{tikzcd}$$

clearly commutes. Finally, A is manifestly final due to being the subset of a product and inheriting the canonical projections. ■